

Confidence-Calibrated Biophysical Geometric Neural Networks for Robust Protein-Ligand Binding-Site Prediction

Research Proposal

Core theme: mathematical biophysical priors + uncertainty-aware geometric reasoning

Prepared from the current PBSP architecture, pilot experiments, and the combined confidence-aware GNN proposal.

1. Summary

This research proposes a confidence-calibrated, biophysically informed geometric graph neural network (GNN) for residue-level protein-ligand binding-site prediction. The central premise is that modern structure-based predictors should not treat every geometric observation as equally reliable. Predicted protein structures contain spatially heterogeneous uncertainty, and this uncertainty can affect distances, orientations, cavity descriptors, and message-passing edges. The proposed method therefore separates sequence/evolutionary evidence from coordinate-dependent geometric evidence and uses calibrated structural confidence to regulate only the geometric component.

The second pillar of the proposal is interpretable mathematical feature engineering. Rather than relying exclusively on latent embeddings, the model incorporates compact biophysical descriptors such as relative solvent accessibility (RSA), side-chain exposure, cavity depth, local packing density, conservation-exposure interactions, and periodic backbone representations. These features are treated as testable inductive priors: they are retained only if controlled ablations demonstrate value beyond ESM-based and geometric representations.

Protein sequence + biophysical priors + calibrated structural uncertainty → robust binding-site prediction

2. Background and Motivation

Protein binding-site prediction is central to structure-based drug discovery, functional annotation, docking, and protein design. Classical pocket-detection methods use surface geometry and cavity statistics, while recent deep-learning methods increasingly combine protein language models with graph or equivariant geometric networks. The rapid growth of predicted structural databases has made structure-based prediction possible for proteins lacking experimentally solved structures.

However, predicted structures are not uniformly reliable. Per-residue confidence values such as pLDDT describe local model confidence, while pairwise uncertainty measures such as Predicted Aligned Error (PAE) describe uncertainty in relative residue or domain placement. A conventional GNN generally propagates messages over spatial edges without distinguishing between highly reliable and uncertain geometry. This creates a potential failure mode: uncertain coordinates may alter local neighborhoods, cavity descriptors, edge distances, and orientations, yet are processed as if they were equally trustworthy.

Naively suppressing low-confidence residues is also undesirable because flexible loops and conformationally dynamic regions can participate directly in binding. The proposed research therefore does not discard uncertain residues. Instead, it preserves sequence and biochemical evidence while selectively attenuating only coordinate-dependent information.

3. Research Gap and Central Hypothesis

The key research gap is not merely whether pLDDT can be added to a binding-site model, but how uncertainty should be represented and where it should act inside the geometric reasoning process. Existing multimodal predictors already combine sequence and structure, and recent work has begun to use confidence-aware adaptation. A stronger research question is therefore whether structural confidence can be calibrated against observed geometric error and then used as an interpretable reliability variable inside message passing and mathematical pocket descriptors.

Central hypothesis: Explicitly separating biochemical evidence from uncertain geometric evidence, while calibrating geometric trust using structural-confidence signals, will improve robustness on predicted and structurally perturbed proteins without unnecessarily suppressing useful sequence information.

4. Research Objectives

1. Develop a multimodal residue representation combining protein language-model embeddings, evolutionary information, and interpretable biophysical descriptors.
2. Develop geometry-specific confidence gating so that uncertainty modulates coordinate-dependent evidence rather than all residue information.
3. Calibrate pLDDT/PAE-derived confidence against observed PDB-to-predicted-structure geometric disagreement.
4. Introduce uncertainty-aware mathematical descriptors, including confidence-weighted local packing density.
5. Evaluate whether engineered descriptors contribute beyond ESM and geometric representations using strict ablation studies.
6. Evaluate robustness using sequence-clustered real-protein benchmarks, paired experimental/predicted structures, and controlled coordinate perturbations.

5. Proposed Protein Representation

Each protein is represented as a residue graph $G = (V, E)$, where each node corresponds to one amino-acid residue and edges connect spatial neighbors under a k -nearest-neighbor and distance-cutoff rule. The architecture uses separate feature branches so that sequence, biophysical, and structural information remain semantically distinguishable before fusion.

5.1 Sequence and Evolutionary Branch

$$h_i^{seq} = f_{seq}(ESM_i, PSSM_i, Conservation_i, Hydrophobicity_i)$$

This branch contains information that does not depend directly on predicted coordinates and therefore should not be automatically suppressed when structural confidence is low.

5.2 Mathematical Biophysical Branch

The biophysical scalar block uses compact, interpretable descriptors. Redundancy screening is used to remove obvious duplicate encodings, but VIF is treated as a diagnostic rather than a hard requirement for the neural model. Final retention is determined by validation and test performance.

Feature	Mathematical form	Interpretation
RSA	$SASA_i / \text{MaxSASA}(\text{AA}_i)$	Residue exposure normalized by amino-acid size.
Side-chain presentation	$SASA_{sc,i} / (SASA_{total,i} + \epsilon)$	Side-chain orientation and presentation at the surface.
Log cavity accessibility	$\log(1 + \text{Depth}_i / (RSA_i + \epsilon))$	Combines concavity/depth with exposure while damping extreme

Feature	Mathematical form	Interpretation
		ratios.
Local packing density	$\sum_j \exp(-d_{ij}^2/(2\sigma^2))$	Local geometric packing around a residue.
Centered conservation-exposure interaction	$(\text{Cons}_i - \mu_{\text{cons}})(\text{RSA}_i - \mu_{\text{RSA}})$	Explicit second-order evolutionary/exposure interaction.
Backbone torsions	$[\sin \varphi_i, \cos \varphi_i, \sin \psi_i, \cos \psi_i]$	Continuous representation of periodic dihedral angles.

5.3 New Uncertainty-Aware Mathematical Feature

The proposal introduces a confidence-weighted local packing density that explicitly separates apparent geometry from geometry supported by reliable coordinates.

$$\begin{aligned}\rho_i &= \sum_{j \in N(i)} \exp(-d_{ij}^2/(2\sigma^2)) \\ \rho_i^{\text{conf}} &= \sum_{j \in N(i)} g_{ij} \exp(-d_{ij}^2/(2\sigma^2)) \\ \Delta\rho_i &= \rho_i - \rho_i^{\text{conf}}\end{aligned}$$

If $\Delta\rho_i$ is close to zero, most of the local packing signal is supported by reliable coordinates. A large $\Delta\rho_i$ indicates that the apparent packing pattern depends substantially on uncertain neighbors. This makes $\Delta\rho$ an interpretable candidate feature for detecting unreliable pocket geometry.

6. Confidence Calibration

Raw pLDDT should not automatically be interpreted as a linear probability of geometric correctness. The proposed model therefore first calibrates confidence against observed geometric disagreement using proteins for which both experimental and predicted structures are available.

$$e_i = ||x_i^{\text{pred}} - x_i^{\text{PDB}}||_2$$

After suitable structural or local alignment, a monotonic calibration model (for example isotonic regression or a monotonic spline) estimates expected structural error as a function of confidence:

$$\begin{aligned}\hat{e}_i &= f_{\text{cal}}(\text{pLDDT}_i) \\ r_i &= \exp(-\hat{e}_i^2/(2s^2))\end{aligned}$$

The value r_i is then used as an empirical residue reliability estimate rather than treating pLDDT/100 as a direct probability.

6.1 Pairwise Reliability Using PAE

$$\begin{aligned}A_{ij} &= (\text{PAE}_{ij} + \text{PAE}_{ji})/2 \\ g_{ij} &= [\min(r_i, r_j)]^\alpha \exp(-A_{ij}/\tau)\end{aligned}$$

This combines local residue reliability with uncertainty in relative residue placement. Parameters α and τ are selected on validation data or learned under constrained parameterization.

7. Modality-Selective Confidence-Gated GNN

The most important architectural choice is that uncertainty gates only coordinate-dependent messages. Low structural confidence does not imply that the residue identity, protein-language representation, conservation profile, or hydrophobicity is unreliable.

$$\begin{aligned}m_{ij} &= m_{ij}^{\text{nonge}} + g_{ij} m_{ij}^{\text{geo}} \\m_{ij}^{\text{nonge}} &= f_N(h_i, h_j, \text{sequence/evolutionary features}) \\m_{ij}^{\text{geo}} &= f_G(h_i, h_j, \text{distance RBFs, local orientation, geometric edge features}) \\h_i^{l+1} &= h_i^l + f_U(h_i^l, \sum_j m_{ij}^l) \\P(y_i=1) &= \sigma(f_{\text{out}}(h_i^L))\end{aligned}$$

Coordinate-derived scalar features are treated similarly. Instead of replacing a structural descriptor s_i with only $r_i s_i$, the model receives $[s_i, r_i, r_i s_i]$. This lets the network distinguish a strong pocket-like signal from the confidence assigned to that signal.

7.1 Interpretable Learned Gate

An unconstrained MLP gate is included as an ablation, but the primary learned gate should preserve a monotonic interpretation: increasing residue reliability should not reduce the edge weight, and increasing pairwise uncertainty should not increase it.

$$\begin{aligned}g_{ij} &= \sigma(\beta_0 + \beta_1 \text{logit}(r_i) + \beta_2 \text{logit}(r_j) - \beta_3 A_{ij}/A_0) \\ \beta_k &= \text{softplus}(\theta_k), \quad k \in \{1, 2, 3\}\end{aligned}$$

8. Research Questions

RQ1: Do explicit mathematical/biophysical features add predictive information beyond ESM and geometric representations?

RQ2: Does confidence-aware geometric message passing outperform ordinary geometry on predicted structures?

RQ3: Does pairwise PAE provide useful information beyond endpoint pLDDT alone?

RQ4: Does empirical confidence calibration outperform raw confidence scaling?

RQ5: Do confidence-weighted mathematical descriptors improve robustness to coordinate perturbation?

9. Dataset and Splitting Protocol

The pilot experiments are treated only as implementation and numerical-feasibility checks. The scientific evaluation will use real protein-ligand complexes from established resources such as BioLiP, PDBbind, COACH420, and HOLO4K, with a matched subset of predicted structures when available.

- Define residue-level binding labels using heavy-atom proximity to a valid non-solvent ligand, using a fixed distance threshold (for example 4.5 Å).
- Cluster protein sequences using MMseqs2 or an equivalent method at a stringent sequence-identity threshold (approximately 30%) and split entire clusters, not individual proteins.

- Fit scalar preprocessing, feature-selection rules, calibration functions, and thresholds strictly on training/validation partitions.
- Maintain a paired PDB-to-predicted-structure test subset to quantify the structure-domain gap directly.

10. Experimental Design

10.1 Controlled Model Ladder

Model	Configuration	Question answered
M0	ESM/PSSM + ordinary geometric GNN	Reference model.
M1	M0 + mathematical biophysical features	Do explicit priors add value?
M2	M1 + minimum-pLDDT geometry gate	Does simple confidence gating help?
M3	M1 + pLDDT/PAE pairwise gate	Does pairwise uncertainty improve gating?
M4	M1 + calibrated confidence + PAE gate	Does empirical confidence calibration help?
M5	M4 + confidence-aware mathematical descriptors	Do uncertainty-weighted priors add further robustness?

10.2 Gating Ablations

- Ungated: $g_{ij} = 1$
- Arithmetic confidence gate
- Minimum endpoint confidence gate
- Multiplicative confidence gate
- pLDDT + PAE analytic gate
- Monotonic learned gate
- Unconstrained MLP gate (ablation only)

10.3 PDB-to-Predicted-Structure Transfer

For the same protein sequence, evaluate the model on an experimental structure and its predicted counterpart while keeping the biological binding labels fixed. The primary robustness quantity is the change in performance caused by replacing experimental coordinates with predicted coordinates.

$$\Delta PR_i = PR-AUC(PDB_i) - PR-AUC(Predicted_i)$$

A robust uncertainty-aware model should reduce the magnitude of this degradation relative to the ordinary GNN.

10.4 Controlled Coordinate Corruption

To test the causal effect of structural uncertainty, experimental structures will also be perturbed under controlled noise.

$$x_i' = x_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma_i^2 I)$$

Noise levels can be uniform for controlled stress testing or confidence-conditioned using the calibrated error model. Robustness is summarized by performance curves across perturbation levels, rather than by a single arbitrary perturbation.

$$R = \int_0^{\sigma_{\max}} PR-AUC(\sigma) d\sigma$$

11. Training Strategy

Training will use protein-level mini-batches and class-imbalance-aware objectives. Focal loss can be used as a baseline objective, but loss choice will be validated rather than assumed optimal. Early stopping and model selection will be based on validation PR-AUC, not training loss.

- Primary optimizer: AdamW with learning-rate scheduling and weight decay.
- Gradient clipping and mixed precision for stable geometric-network training.
- Frozen ESM embeddings and LoRA fine-tuning treated as separate experimental configurations; precomputed ESM tensors cannot be LoRA-fine-tuned during downstream training.
- Hyperparameters selected only on training/validation data.
- Multiple random seeds reported for each major architecture.

12. Evaluation Framework

Because positive binding residues are sparse, PR-AUC is the primary residue-level metric. ROC-AUC is reported as a secondary ranking metric. Threshold-dependent metrics use a decision threshold selected on the validation partition rather than a fixed 0.5 cutoff.

- PR-AUC (primary)
- ROC-AUC
- Validation-selected MCC
- Precision, recall, and F1
- Top-K precision / hit rate
- Pocket-level Distance to Center of Action (DCA)
- Performance stratified by confidence bands

- Robustness under controlled coordinate corruption

$$\tau^* = \operatorname{argmax}_{\tau} \operatorname{MCC}(y_{\text{val}}, p_{\text{val}} \geq \tau)$$

Confidence-stratified analysis will report performance across bands such as <50, 50–70, 70–90, and >90 pLDDT. These bands are interpreted as confidence strata, not as direct disorder labels.

13. Feature-Engineering Policy

Feature engineering remains an important part of this proposal, but it is used to encode domain knowledge and improve interpretability rather than to impose linear-model assumptions on the neural network.

- Use priority-guided correlation screening to remove obvious duplicates such as raw SASA versus RSA or multiple nearly equivalent hydrophobicity scales.
- Use VIF and correlation matrices as diagnostics for the scalar block, not as proof that a feature is useful.
- Use centering for interaction terms such as conservation $\times$ RSA, but do not claim that centering guarantees mathematical orthogonality in general.
- Retain a feature only if controlled ablations show reproducible benefit or it serves a clearly justified interpretability role.
- Compare the compact engineered feature block against raw scalar features and against a model with no handcrafted scalar inputs.

14. Expected Contributions

Confidence-calibrated structural reliability: A data-driven mapping from structural-confidence signals to empirical geometric reliability.

Modality-selective uncertainty reasoning: A GNN in which uncertainty regulates coordinate-dependent messages without discarding sequence/evolutionary evidence.

Uncertainty-aware mathematical pocket descriptors: Confidence-weighted packing and cavity-related descriptors with direct biological interpretation.

Controlled robustness benchmark: Paired experimental/predicted structures plus calibrated coordinate corruption for measuring sensitivity to geometry.

Rigorous feature-engineering analysis: Ablations establishing when explicit biophysical priors complement modern protein embeddings and geometric networks.

15. Risks and Mitigations

Risk	Mitigation
Gating effect is small or statistically insignificant	Treat this as a scientifically valid result; the controlled robustness analysis remains informative.
Low-confidence regions contain real binding residues	Gate only geometric evidence; preserve sequence/evolutionary information and perform

Risk	Mitigation
	confidence-stratified recall analysis.
Model overfits a moderate-size benchmark	Use cluster-level splits, regularization, reduced ESM dimensionality, early stopping, and repeated seeds.
pLDDT is confused with disorder	Treat pLDDT as confidence only; use explicit disorder predictors as sensitivity analysis if needed.
Feature engineering becomes overly complex	Require ablation evidence for each engineered descriptor and keep the final scalar block compact.

17. Expected Outcome and Significance

The expected outcome is a residue-level and pocket-level predictor that combines modern sequence representations with interpretable biophysical priors while explicitly representing uncertainty in structural geometry. The objective is not simply to improve a benchmark score, but to determine how a structure-aware model should reason when its coordinates are imperfect.

How should a geometric neural network reason when it does not fully trust its geometry?

If successful, the resulting framework could generalize beyond ligand-pocket prediction to other structural-bioinformatics tasks in which predicted geometry is useful but uncertain, including protein interfaces, functional-site prediction, and structure-based molecular design.

18. Selected Background References

- AlphaFold Protein Structure Database documentation and confidence-score guidance (pLDDT and PAE).
- Tien et al. (2013). Maximum allowed solvent accessibilities of residues in proteins.
- Kyte and Doolittle. Hydropathy scale for protein sequence analysis.
- P2Rank / PrankWeb literature for geometry-based ligand-binding-site prediction.
- Recent multimodal and geometric protein binding-site prediction literature, including ProtGeoNet-Pocket and related graph/equivariant approaches.
- Recent confidence-aware PDB-to-AlphaFold binding-site prediction literature, including ProtCross.